STAT4601-Assignment2

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Question 1 - PCA

```
library(stats)
library(here)
```

```
## Warning: package 'here' was built under R version 4.3.3

## here() starts at C:/Users/tdthe/Downloads/My R files/STAT4601/Assignment 2
```

```
library(fastICA)
```

```
## Warning: package 'fastICA' was built under R version 4.3.3
```

```
library(tidyverse)
```

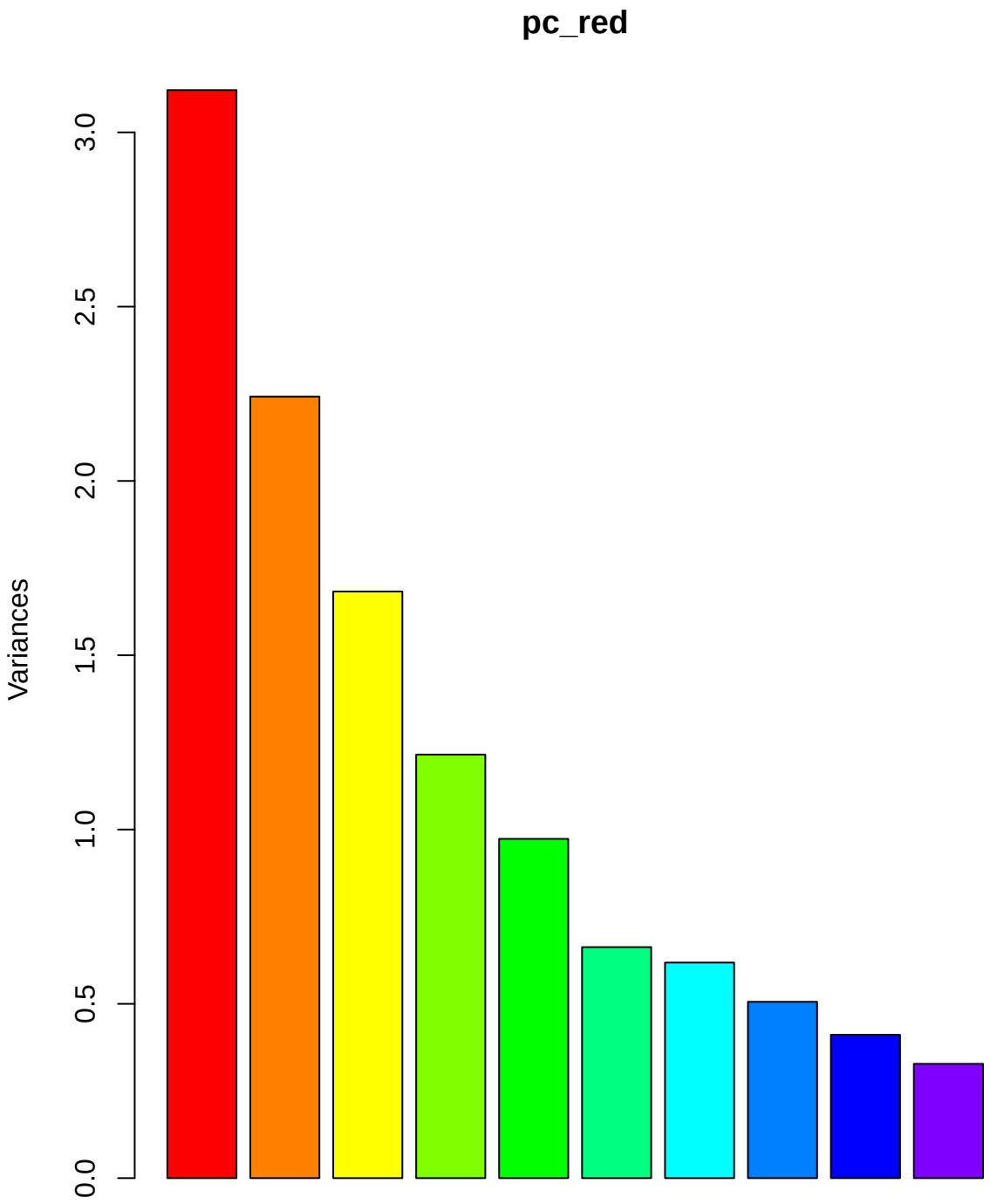
```
## Warning: package 'tidyverse' was built under R version 4.3.3

## Warning: package 'ggplot2' was built under R version 4.3.3

## Warning: package 'dplyr' was built under R version 4.3.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
set.seed(123)

fpath<-file.path("C:","Users","tdthe","Downloads","My R files","STAT4601","Assignment 2", .Platform$file
paste_red<-paste(fpath,"winequality-red.csv",sep="")
raw_data_red<-read.csv(paste_red)
```

```r
categories<-c("Fixed Acidity","Volatile Acidity", "Citric Acid", "Residual Sugar", "Chlorides", "Free Su
dataset_red<-separate(raw_data_red,col=everything(),into=categories, sep=";")
red_data<-as.data.frame(lapply(dataset_red, as.numeric))
pc_red<-prcomp(red_data, scale.=TRUE) #the scaled PCA
summary(pc_red)
```

```
## Importance of components:
##                           PC1    PC2    PC3    PC4     PC5     PC6     PC7
## Standard deviation     1.7667 1.4973 1.2973 1.1023 0.98654 0.81400 0.78633
## Proportion of Variance 0.2601 0.1868 0.1402 0.1013 0.08111 0.05522 0.05153
## Cumulative Proportion  0.2601 0.4469 0.5872 0.6884 0.76952 0.82474 0.87626
##                            PC8     PC9    PC10    PC11    PC12
## Standard deviation     0.71125 0.64133 0.57264 0.42452 0.24396
## Proportion of Variance 0.04216 0.03428 0.02733 0.01502 0.00496
## Cumulative Proportion  0.91842 0.95270 0.98002 0.99504 1.00000
```

```r
plot(pc_red,col=rainbow(12))
```



Looking at the contribution to the variance for each Principal Component, PC1 contributes to 26.01% of the variance, and PC2 adds 18.68% of it, PC3 adds 14.02%, and PC4 adds 10.13%. Everything else (from PC5 to PC12) contributes less than 10% of the variance. As such, it wouldn't add a significant amount of

information to the variance of the data set, compared to the other components.

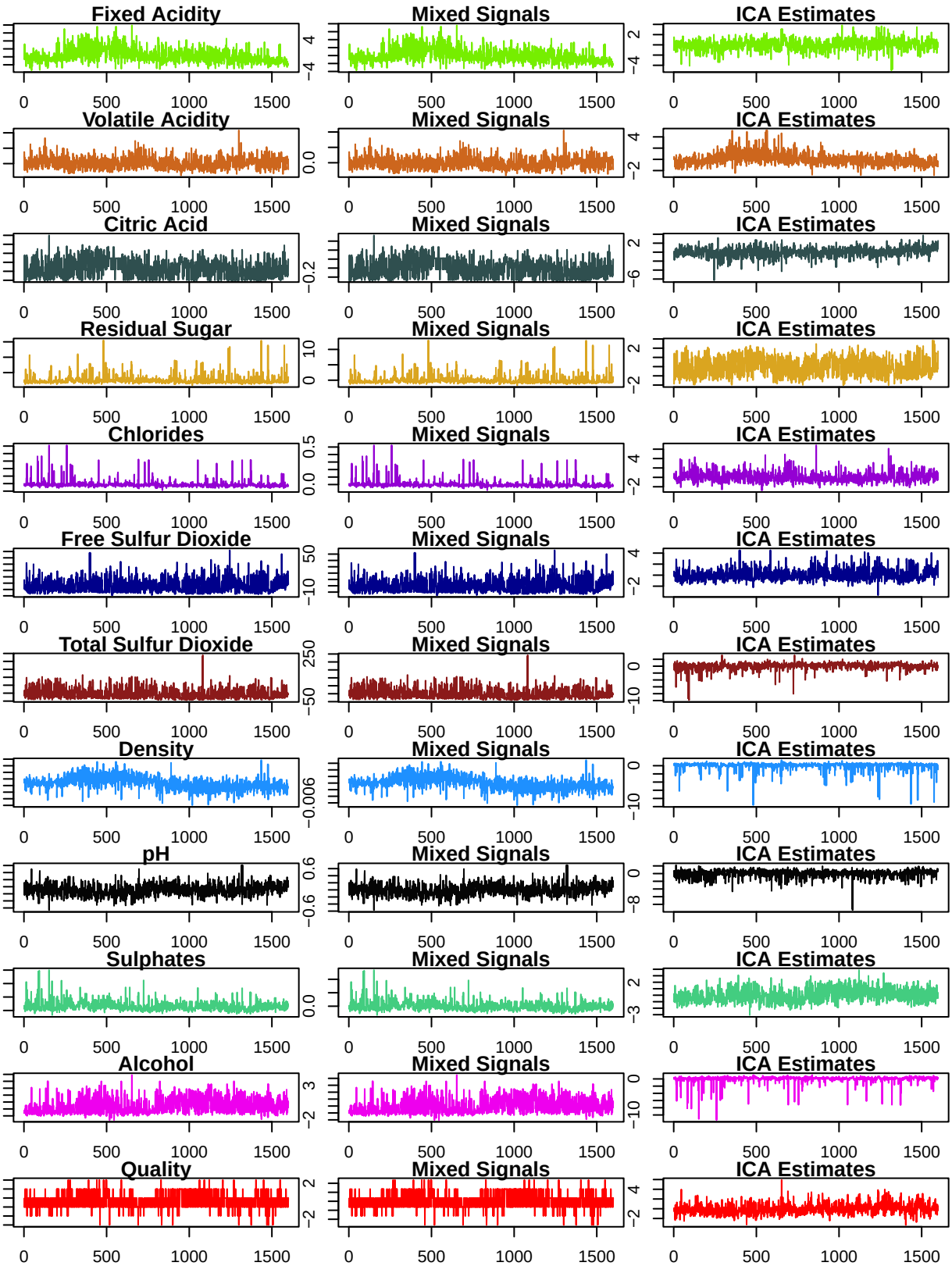
Question 1 - ICA

```
set.seed(123)
ic_red<-fastICA(red_data,12, verbose=TRUE)
```

```
## Centering

## Whitening

## Symmetric FastICA using logcosh approx. to neg-entropy function

## Iteration 1 tol = 0.9281121

## Iteration 2 tol = 0.2183723

## Iteration 3 tol = 0.08195308

## Iteration 4 tol = 0.03750487

## Iteration 5 tol = 0.02186904

## Iteration 6 tol = 0.006393529

## Iteration 7 tol = 0.005834411

## Iteration 8 tol = 0.002232956

## Iteration 9 tol = 0.001233711

## Iteration 10 tol = 0.0006576378

## Iteration 11 tol = 0.0003872945

## Iteration 12 tol = 0.0002448841

## Iteration 13 tol = 0.0001486868

## Iteration 14 tol = 9.843443e-05
```

```
colorVec <- c("chartreuse2", "chocolate3", "darkslategray","goldenrod", "darkviolet", "darkblue", "firel

oldpar <- par(mfcol = c(12, 3), mar=c(1.85, 1, 1, 1)) #to chart the ica estimates, mixed signals, and t
for(i in 1:1){
   for(j in 1:12){
      plot(1:1599, red_data[, j], type = "l", main = categories[j],
           xlab = "", ylab = "",
           lwd = 1, col = colorVec[j])
```

```r
  }
  for(j in 1:12){
     plot(1:1599, ic_red$X[, j], type = "l", main = "Mixed Signals",
          xlab = "", ylab = "",
          lwd = 1, col = colorVec[j])
  }
  for(j in 1:12){
     plot(1:1599, ic_red$S[, j], type = "l", main = "ICA Estimates",
          xlab = "", ylab = "",
          lwd = 1, col = colorVec[j])
  }
}
```



For the ICA, the estimates are fairly close, though mixed up, and flipped upside down. Even though there's a lot of data with a lot of mixed signals, it was able to differentiate them to be fairly accurate. Similar to PCA, the first few iterations yielded a large percentage, while the others aren't as large. The First iteration

got about 92.8% of the estimates, while 21.8% were obtained by iteration 2. The other iterations yielded less than 10%

Question 2

Since the file seems to be too great for both Gower and hierarchical clustering to process and comprehend in a neat format, I will use a sample of the data given proportional to the .

```
library(dendextend)
```

```
## Warning: package 'dendextend' was built under R version 4.3.3

##
## ---------------------
## Welcome to dendextend version 1.19.0
## Type citation('dendextend') for how to cite the package.
##
## Type browseVignettes(package = 'dendextend') for the package vignette.
## The github page is: https://github.com/talgalili/dendextend/
##
## Suggestions and bug-reports can be submitted at: https://github.com/talgalili/dendextend/issues
## You may ask questions at stackoverflow, use the r and dendextend tags:
##   https://stackoverflow.com/questions/tagged/dendextend
##
##  To suppress this message use:  suppressPackageStartupMessages(library(dendextend))
## ---------------------

##
## Attaching package: 'dendextend'

## The following object is masked from 'package:stats':
##
##     cutree
```

```
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.3.3
```

```
library(ggplot2)
library(dplyr)
set.seed(123)

fpath2<-file.path("C:","Users","tdthe","Downloads","My R files","STAT4601","Assignment 2","diabetes_012_
dia_name<-paste(fpath2,"diabetes_012_health_indicators_BRFSS2015.csv",sep="")
raw_diab<-read.csv(dia_name)

raw_diab<-raw_diab|>
  mutate(Diabetes_012<-as.factor(Diabetes_012))

prop.table(table(raw_diab$Diabetes_012)) #to ensure that the proportion of the factor levels are around
```

```
##
##          0          1          2
## 0.84241170 0.01825528 0.13933302
```

```r
sample_size <- 0.0004

# This is to sample the data to get 10,000 observations as well as proportional to the factor levels
rd_sample <- raw_diab %>%
  group_by(Diabetes_012) %>%
  sample_frac(sample_size)

prop.table(table(rd_sample$Diabetes_012)) #To ensure that the sample have similar proportions to the fu
```

```
##
##          0          1          2
## 0.84158416 0.01980198 0.13861386
```

```r
g_diab<-daisy(rd_sample, metric ="gower") #Gower's Distance
```

```
## Warning in daisy(rd_sample, metric = "gower"): binary variable(s) 2, 3, 4, 6,
## 7, 8, 9, 10, 11, 12, 13, 14, 18, 19 treated as interval scaled
```

```r
gd_mat<-as.matrix(g_diab)
#Most Similar in the sample
raw_diab[
which(gd_mat == min(gd_mat[gd_mat != min(gd_mat)]),
arr.ind = TRUE)[1, ], ]
```

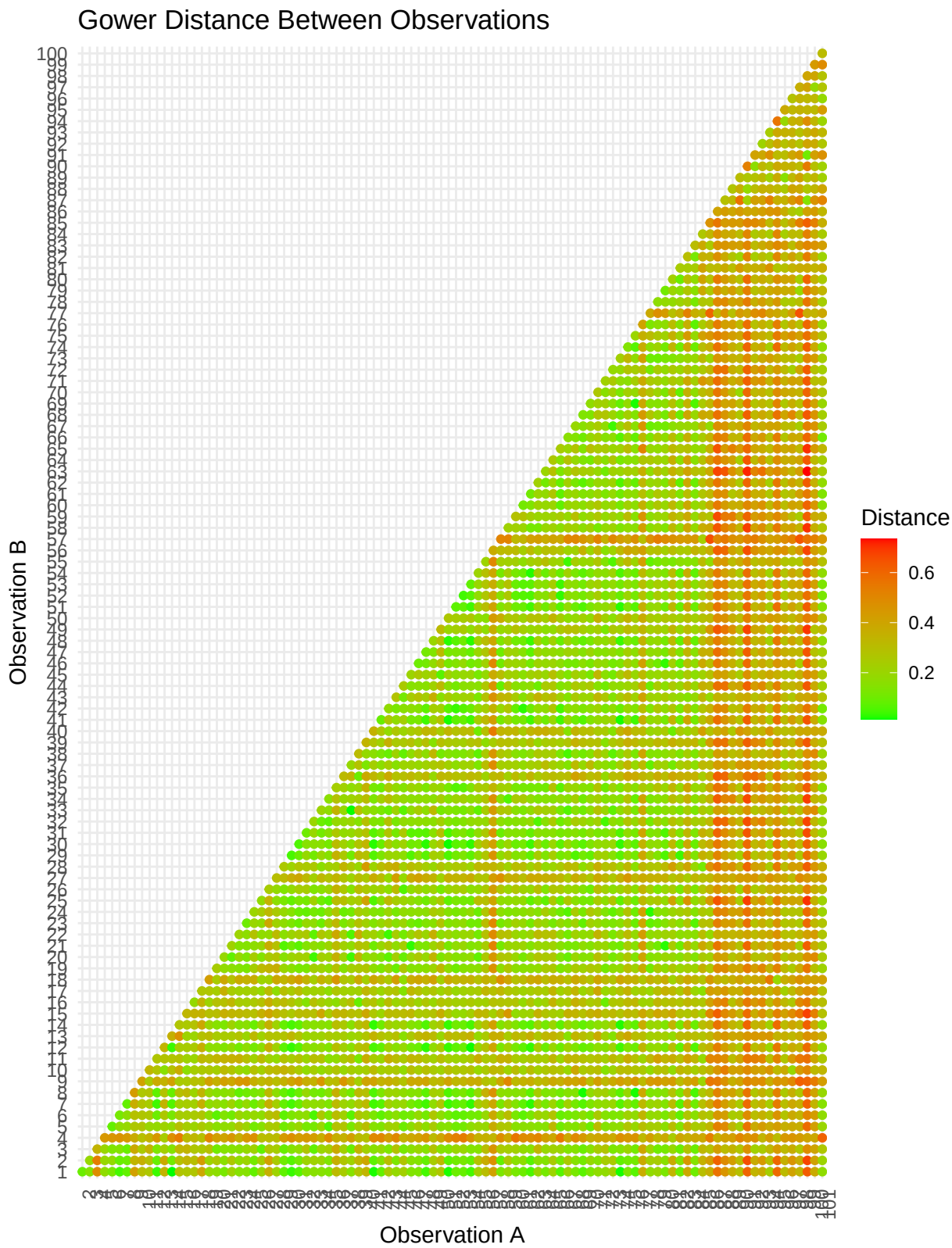
```
##    Diabetes_012 HighBP HighChol CholCheck BMI Smoker Stroke
## 14            2      1        1         1  28      0      0
## 1             0      1        1         1  40      1      0
##    HeartDiseaseorAttack PhysActivity Fruits Veggies HvyAlcoholConsump
## 14                    0            0      0       1                 0
## 1                     0            0      0       1                 0
##    AnyHealthcare NoDocbcCost GenHlth MentHlth PhysHlth DiffWalk Sex Age
## 14             1           0       4        0        0        1   0  11
## 1              1           0       5       18       15        1   0   9
##    Education Income Diabetes_012 <- as.factor(Diabetes_012)
## 14         4      6                                       2
## 1          4      3                                       0
```

```r
#Least Similar in the sample
raw_diab[
which(gd_mat == max(gd_mat[gd_mat != max(gd_mat)]),
arr.ind = TRUE)[1, ], ]
```

```
##    Diabetes_012 HighBP HighChol CholCheck BMI Smoker Stroke
## 91            0      1        1         1  32      1      0
## 63            0      1        0         1  34      1      0
##    HeartDiseaseorAttack PhysActivity Fruits Veggies HvyAlcoholConsump
## 91                    0            0      0       1                 0
## 63                    0            1      1       1                 0
##    AnyHealthcare NoDocbcCost GenHlth MentHlth PhysHlth DiffWalk Sex Age
## 91             1           0       4        5       15        0   0  10
## 63             1           1       3        0        0        0   0   9
```

```
##    Education Income Diabetes_012 <- as.factor(Diabetes_012)
## 91         5      8                                       0
## 63         6      5                                       0
```

```r
gd_mat[upper.tri(gd_mat, diag = TRUE)] <- NA
gower_diab <- as.data.frame(as.table(gd_mat))
gower_diab <- na.omit(gower_diab)
colnames(gower_diab) <- c("ObservationA", "ObservationB", "Distance")
#Heatmap of the distance between each observation in the sample
ggplot(gower_diab, aes(x = ObservationA, y = ObservationB, color = Distance)) +
geom_point() +
scale_color_gradient(low = "green", high = "red") +
labs(title = "Gower Distance Between Observations",
x = "Observation A", y = "Observation B", color = "Distance") +
theme_minimal() + theme(axis.text.x = element_text(angle = 90, hjust = 1))
```

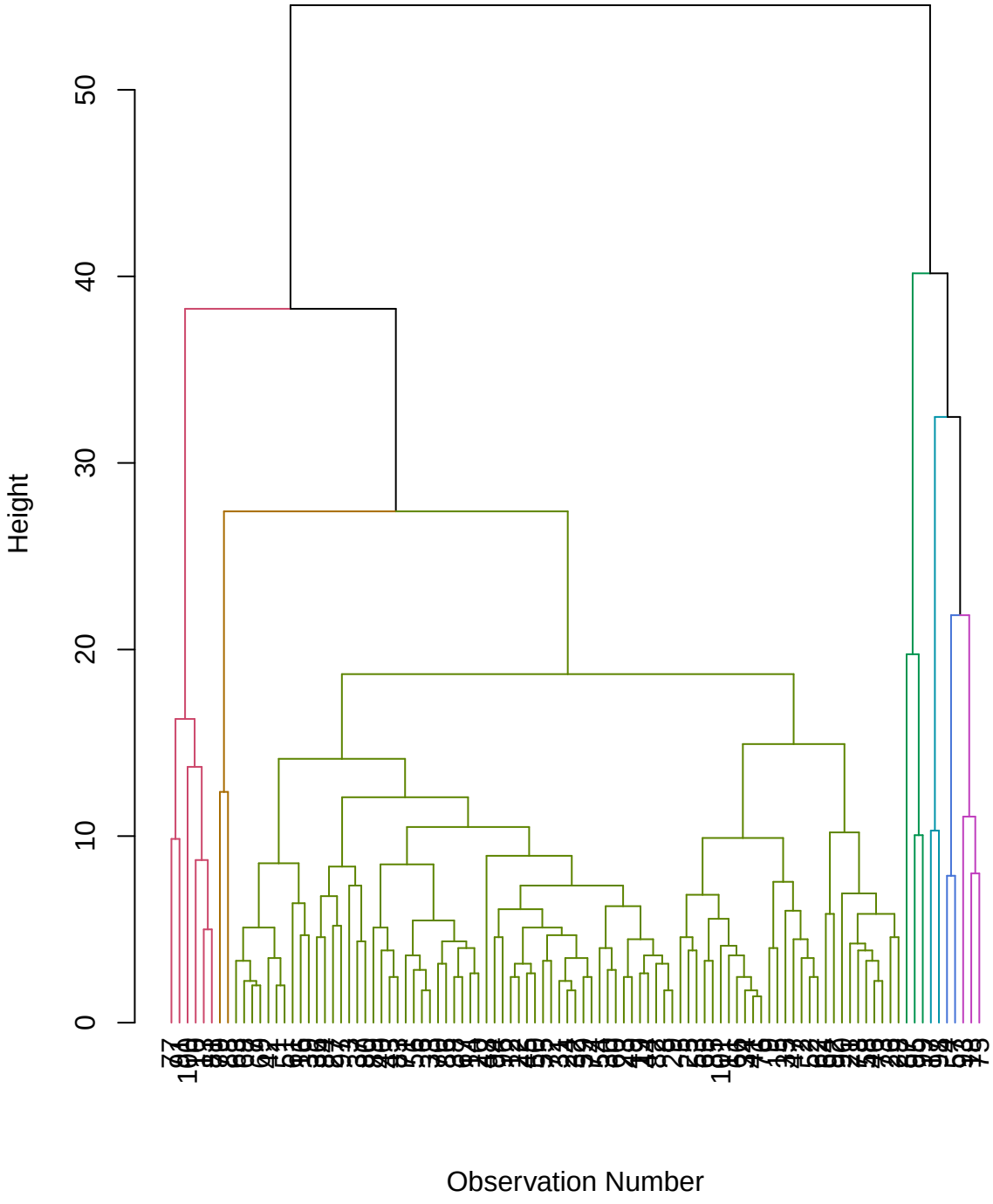


The following output is the most and least similar of the sample taken, as well as a heat map of the sample and their differences with each other, using the Gower's distance method.It shows that there is a group that are fairly similar, but a lot of them have some distance. There are some data that are really different, which

would be indicated by red, but not as much as green.

In this sample, observation 14 and 1 are the most similar, while observation 91 and 63 are the least similar

```
set.seed(123)


d_dist<-dist(rd_sample)
h_diab<-hclust(d_dist, method="complete") #Hierarchical Clustering
h_dend<-as.dendrogram(h_diab)
plot(colour_branches(h_dend, k=7), xlab="Observation Number", ylab="Height") #Visualization with 7 colo
```



The following output is a dendrogram of a hierarchical clustering method.A large part of the sample are very similar, that it would consider putting them with each other in a cluster. There are one or two that is considered by itself for quite some time before joining with a another datapoint, which also reflects to that

there isn't as much data that are really different from the Gower's distance.